

SOF NSO Science — Topic Deep-Dive: Water & Air

For Ambar, Class III CBSE

1. Sources of water

Water comes from many places: - **Rain** — the main source of fresh water. - **Rivers, lakes, ponds, seas** — surface water. - **Wells, hand pumps, borewells** — underground water. - **Taps** at home bring us clean, stored water.

We should always drink **clean, covered water** so that germs and dirt do not get in.

2. States of water

Water is found in all **three states**:

State	Form of water	Example
Solid	Ice	Ice cubes, snow
Liquid	Water	Water we drink, rain
Gas	Water vapour (steam)	Steam from boiling water

Water can change from one state to another when we heat or cool it.

3. The water cycle

The Sun's heat moves water round and round the Earth. This is the **water cycle**.

1. **Evaporation** — The Sun heats water in seas, rivers and ponds. The water turns into **water vapour** (a gas) and rises up.
2. **Condensation** — High up, the vapour **cools** and forms tiny drops that make **clouds**.
3. **Rain (precipitation)** — The drops join, become heavy, and fall down as **rain**.
4. **Collection** — Rain water collects in rivers, lakes and seas, and the cycle begins again.

Everyday proof: - Wet clothes dry in the sun because water **evaporates**. - Water drops appear on a cold glass because vapour in the air **condenses** on it.

4. Saving water

Clean water is precious and limited. We should save it: - Close the tap while brushing teeth or soaping hands. - Fix leaking taps and pipes. - Reuse water (like vegetable-washing water for plants). - Do not let tanks overflow.

5. Air is everywhere

Air is all around us, even though we cannot see it. We know air is there when: - the wind moves leaves and flies a kite, - a balloon becomes big when filled with air.

Air **takes up space** and can be felt as **wind** (moving air). Wind can turn windmills and push sailing boats.

What is in the air

Gas	Its use
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Oxygen

We (and animals) **breathe it in** to stay alive

Carbon dioxide

Plants take it in to make their food

We breathe **in oxygen** and breathe **out carbon dioxide**. Plants take in the carbon dioxide and give out oxygen — so plants and animals help each other.

6. Air pollution basics

Air pollution is when the air becomes dirty and smoky. It is caused by: - smoke from vehicles and factories, - burning rubbish and coal.

We can keep the air clean by **planting more trees**, using cycles and buses, and not burning waste. Trees clean the air by taking in carbon dioxide and giving out oxygen.

Common traps

The trap

The truth

"We breathe in carbon dioxide"

We breathe **in oxygen**, out carbon dioxide

"Plants take in oxygen to make food"

Plants take in **carbon dioxide** to make food

"Rain comes up from the ground"

Rain **falls from clouds** in the sky

"Ice is the gas form of water"

Ice is the **solid** form; steam is the gas form

"Trees pollute the air"

Trees **clean** the air; smoke pollutes it

"Clouds are made of smoke"

Clouds are made of tiny **water drops**

High-frequency question types

1. **Water cycle order:** evaporation → condensation → rain.
 2. **State of water:** "Ice is the ___ form of water" → solid.
 3. **Gas we breathe:** "We breathe in ___" → oxygen.
 4. **Gas plants use:** "Plants take in ___" → carbon dioxide.
 5. **Air present:** "A kite flies because of ___" → wind.
 6. **Saving water:** "Which saves water?" → closing the tap while brushing.
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Key facts checklist

- ☐ Sources of water: rain, rivers, lakes, wells, taps.
 - ☐ Water has three states: ice (solid), water (liquid), vapour (gas).
 - ☐ Water cycle: evaporation → condensation → rain → collection.
 - ☐ The Sun's heat drives the water cycle.
 - ☐ We breathe in oxygen, breathe out carbon dioxide.
 - ☐ Plants take in carbon dioxide to make food, give out oxygen.
 - ☐ Moving air is wind; air is everywhere and takes up space.
 - ☐ Save water; plant trees to keep air clean.
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Quick drill (10 questions)

1. Name one natural source of water.
2. Ice is the ___ form of water.
3. What is water vapour — solid, liquid or gas?
4. What is it called when water turns into vapour on heating?

5. What is it called when vapour cools into drops?
6. Water falling from clouds is called ____.
7. Which gas do we breathe in?
8. Which gas do plants take in to make food?
9. Moving air is called ____.
10. Name one way to keep the air clean.

Answers: 1. River/lake/rain (any) 2. Solid 3. Gas 4. Evaporation 5. Condensation 6. Rain 7. Oxygen 8. Carbon dioxide 9. Wind 10. Plant more trees